Capstone

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Capstone project

Analysis for bike-share company

A bike-share program that features more than 5,800 bicycles and 600 docking stations. Cyclistic sets itself apart by also offering reclining bikes, hand tricycles, and cargo bikes, making bike-share more inclusive to people with disabilities and riders who can not use a standard two-wheeled bike. Pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members.

Goal: Design marketing strategies aimed at converting casual riders into annual members.

Ask

How do annual members and casual riders use Cyclistic bikes differently?

Task: To analyze the activity and nature of customer trips at 2 types of fares.

Prepare

```
#rm(list = ls()) #clean the environment
```

Download the travel data for the last 12 months, which is all 2022 year.

Download the previous 12 months of Cyclistic trip data: https://divvy-tripdata.s3.amazonaws.com/index.html (https://divvy-tripdata.s3.amazonaws.com/index.html)

Data License Agreement: https://ride.divvybikes.com/data-license-agreement (https://ride.divvybikes.com/data-license-agreement)

The data is organized into files by month and stored in the same directory as the code file for analysis.

Get libraries

```
library(tidyverse)  #helps wrangle data
```

```
## ── Attaching packages ─────────────────────────────────────── tidyverse 1.3.2 ──
## ✔ ggplot2 3.4.0      ✔ purrr   1.0.0
## ✔ tibble  3.1.8      ✔ dplyr   1.0.10
## ✔ tidyr   1.2.1      ✔ stringr 1.5.0
## ✔ readr   2.1.3      ✔ forcats 0.5.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
```

```
library(lubridate)  #helps wrangle date attributes
```

```
## Загрузка требуемого пакета: timechange
##
## Присоединяю пакет: 'lubridate'
##
## Следующие объекты скрыты от 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)  #helps visualize data
```

Get data

```
tripdata_01 <- read.csv(file = '202201-divvy-tripdata.csv')
tripdata_02 <- read.csv(file = '202202-divvy-tripdata.csv')
tripdata_03 <- read.csv(file = '202203-divvy-tripdata.csv')
tripdata_04 <- read.csv(file = '202204-divvy-tripdata.csv')
tripdata_05 <- read.csv(file = '202205-divvy-tripdata.csv')
tripdata_06 <- read.csv(file = '202206-divvy-tripdata.csv')
tripdata_07 <- read.csv(file = '202207-divvy-tripdata.csv')
tripdata_08 <- read.csv(file = '202208-divvy-tripdata.csv')
tripdata_09 <- read.csv(file = '202209-divvy-tripdata.csv')
tripdata_10 <- read.csv(file = '202210-divvy-tripdata.csv')
tripdata_11 <- read.csv(file = '202211-divvy-tripdata.csv')
tripdata_12 <- read.csv(file = '202212-divvy-tripdata.csv')
tripdata <- rbind(tripdata_01, tripdata_02, tripdata_03, tripdata_04, tripdata_05, tripdata_06, tripdata_07, trip
data_08, tripdata_09, tripdata_10, tripdata_11, tripdata_12)
#drop unnecesary dataframes
rm(tripdata_12)
rm(tripdata_11)
rm(tripdata_10)
rm(tripdata_09)
rm(tripdata_08)
rm(tripdata_07)
rm(tripdata_06)
rm(tripdata_05)
rm(tripdata_04)
rm(tripdata_03)
rm(tripdata_02)
rm(tripdata_01)
```

View data

```
head(tripdata)
```

```
##            ride_id rideable_type          started_at            ended_at
## 1 C2F7DD78E82EC875 electric_bike 2022-01-13 11:59:47 2022-01-13 12:02:44
## 2 A6CF8980A652D272 electric_bike 2022-01-10 08:41:56 2022-01-10 08:46:17
## 3 BD0F91DFF741C66D  classic_bike 2022-01-25 04:53:40 2022-01-25 04:58:01
## 4 CBB80ED419105406  classic_bike 2022-01-04 00:18:04 2022-01-04 00:33:00
## 5 DDC963BFDDA51EEA  classic_bike 2022-01-20 01:31:10 2022-01-20 01:37:12
## 6 A39C6F6CC0586C0B  classic_bike 2022-01-11 18:48:09 2022-01-11 18:51:31
##              start_station_name start_station_id              end_station_name
## 1      Glenwood Ave & Touhy Ave              525          Clark St & Touhy Ave
## 2      Glenwood Ave & Touhy Ave              525          Clark St & Touhy Ave
## 3 Sheffield Ave & Fullerton Ave     TA1306000016 Greenview Ave & Fullerton Ave
## 4      Clark St & Bryn Mawr Ave     KA1504000151     Paulina St & Montrose Ave
## 5   Michigan Ave & Jackson Blvd     TA1309000002        State St & Randolph St
## 6         Wood St & Chicago Ave              637       Honore St & Division St
##   end_station_id start_lat start_lng  end_lat   end_lng member_casual
## 1         RP-007  42.01280 -87.66591 42.01256 -87.67437        casual
## 2         RP-007  42.01276 -87.66597 42.01256 -87.67437        casual
## 3   TA1307000001  41.92560 -87.65371 41.92533 -87.66580        member
## 4   TA1309000021  41.98359 -87.66915 41.96151 -87.67139        casual
## 5   TA1305000029  41.87785 -87.62408 41.88462 -87.62783        member
## 6   TA1305000034  41.89563 -87.67207 41.90312 -87.67394        member
```

```
View(tripdata)
```

```
glimpse(tripdata)
```

```
## Rows: 5,667,717
## Columns: 13
## $ ride_id            <chr> "C2F7DD78E82EC875", "A6CF8980A652D272", "BD0F91DFF7…
## $ rideable_type      <chr> "electric_bike", "electric_bike", "classic_bike", "…
## $ started_at         <chr> "2022-01-13 11:59:47", "2022-01-10 08:41:56", "2022…
## $ ended_at           <chr> "2022-01-13 12:02:44", "2022-01-10 08:46:17", "2022…
## $ start_station_name <chr> "Glenwood Ave & Touhy Ave", "Glenwood Ave & Touhy A…
## $ start_station_id   <chr> "525", "525", "TA1306000016", "KA1504000151", "TA13…
## $ end_station_name   <chr> "Clark St & Touhy Ave", "Clark St & Touhy Ave", "Gr…
## $ end_station_id     <chr> "RP-007", "RP-007", "TA1307000001", "TA1309000021",…
## $ start_lat          <dbl> 42.01280, 42.01276, 41.92560, 41.98359, 41.87785, 4…
## $ start_lng          <dbl> -87.66591, -87.66597, -87.65371, -87.66915, -87.624…
## $ end_lat            <dbl> 42.01256, 42.01256, 41.92533, 41.96151, 41.88462, 4…
## $ end_lng            <dbl> -87.67437, -87.67437, -87.66580, -87.67139, -87.627…
## $ member_casual      <chr> "casual", "casual", "member", "casual", "member", "…
```

```
colnames(tripdata)
```

```
##  [1] "ride_id"            "rideable_type"      "started_at"
##  [4] "ended_at"           "start_station_name" "start_station_id"
##  [7] "end_station_name"   "end_station_id"     "start_lat"
## [10] "start_lng"          "end_lat"            "end_lng"
## [13] "member_casual"
```

Process

Get data frame in proper data formats.

Convert ride_id and rideable_type to character so that they can stack correctly

```
tripdata <-  mutate(tripdata, ride_id = as.character(ride_id)
                   ,rideable_type = as.character(rideable_type))
```

Calculate the length of each ride by subtracting the column started_at from the column ended_at.

```
tripdata$ended <- strptime(tripdata$ended_at, "%Y-%m-%d %H:%M:%S", tz = "EST5EDT")
tripdata$started <- strptime(tripdata$started_at, "%Y-%m-%d %H:%M:%S", tz = "EST5EDT")
# Add length of the ride
tripdata$ride_length_min <- difftime(tripdata$ended,tripdata$started, units = "mins")
```

Get month of the ride

```
# Get month of the ride
tripdata$month <- month(tripdata$ended)
```

Get day of the week of the ride, where 1 = Sunday and 7 = Saturday

```
# Get day of the week of the ride, where 1 = Sunday and 7 = Saturday
tripdata$weekday <- wday(tripdata$ended, week_start=1)
```

```
View(tripdata)
```

```
#View statistics
summary(tripdata)
```

```
##    ride_id          rideable_type       started_at          ended_at
##  Length:5667717     Length:5667717     Length:5667717     Length:5667717
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##  start_station_name start_station_id   end_station_name   end_station_id
##  Length:5667717     Length:5667717     Length:5667717     Length:5667717
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##    start_lat       start_lng         end_lat         end_lng
##  Min.   :41.64   Min.   :-87.84   Min.   : 0.00   Min.   :-88.14
##  1st Qu.:41.88   1st Qu.:-87.66   1st Qu.:41.88   1st Qu.:-87.66
##  Median :41.90   Median :-87.64   Median :41.90   Median :-87.64
##  Mean   :41.90   Mean   :-87.65   Mean   :41.90   Mean   :-87.65
##  3rd Qu.:41.93   3rd Qu.:-87.63   3rd Qu.:41.93   3rd Qu.:-87.63
##  Max.   :45.64   Max.   :-73.80   Max.   :42.37   Max.   :  0.00
##                                   NA's   :5858    NA's   :5858
##  member_casual          ended
##  Length:5667717     Min.   :2022-01-01 00:01:48.00
##  Class :character   1st Qu.:2022-05-28 19:43:07.00
##  Mode  :character   Median :2022-07-22 15:24:44.00
##                     Mean   :2022-07-20 07:48:39.74
##                     3rd Qu.:2022-09-16 07:39:03.00
##                     Max.   :2023-01-02 04:56:45.00
##
##     started                        ride_length_min       month
##  Min.   :2022-01-01 00:00:05.00   Length:5667717    Min.   : 1.000
##  1st Qu.:2022-05-28 19:21:05.00   Class :difftime   1st Qu.: 5.000
##  Median :2022-07-22 15:03:59.00   Mode  :numeric    Median : 7.000
##  Mean   :2022-07-20 07:29:13.11                     Mean   : 7.112
##  3rd Qu.:2022-09-16 07:21:29.00                     3rd Qu.: 9.000
##  Max.   :2022-12-31 23:59:26.00                     Max.   :12.000
##
##     weekday
##  Min.   :1.000
##  1st Qu.:2.000
##  Median :4.000
##  Mean   :4.062
##  3rd Qu.:6.000
##  Max.   :7.000
##
```

Delete lines without data, because there are 5858 lines with missing data out of 5667717, which is about 0.1%

```
tripdata <- na.omit(tripdata)
```

Remove “bad” data. The dataframe includes a few hundred entries when bikes were taken out of docks and checked for quality by Divvy or ride_length was negative.

```
tripdata_v2 <- tripdata[!(tripdata$start_station_name == "HQ QR" | tripdata$ride_length<0),]
```

Additional: Calculate trip distance

```
#library(geosphere)
#Calculate the distance
#tripdata$ride_dist <- distm(tripdata[,9:10], tripdata[,11:12])

#tripdata %>%
#  rowwise() %>%
#  mutate(ride_dist = distm(tripdata[,9:10], tripdata[,11:12]))
```

Analyze

```
table(tripdata_v2$member_casual)
```

```
##
##  casual  member
## 2316808 3344951
```

There are more members rides then casual.

Descriptive analysis on ride_length (all figures in mins)

```
mean(tripdata_v2$ride_length) #straight average (total ride length / rides)
```

```
## Time difference of 16.33073 mins
```

```
median(tripdata_v2$ride_length) #midpoint number in the ascending array of ride lengths
```

```
## Time difference of 10.26667 mins
```

```
max(tripdata_v2$ride_length) #longest ride
```

```
## Time difference of 34294.07 mins
```

```
min(tripdata_v2$ride_length) #shortest ride
```

```
## Time difference of 0 mins
```

Compare members and casual users by ride time.

```
aggregate(tripdata_v2$ride_length ~ tripdata_v2$member_casual, FUN = mean)
```

```
##   tripdata_v2$member_casual tripdata_v2$ride_length
## 1                    casual           21.98804 mins
## 2                    member           12.41231 mins
```

```
aggregate(tripdata_v2$ride_length ~ tripdata_v2$member_casual, FUN = median)
```

```
##   tripdata_v2$member_casual tripdata_v2$ride_length
## 1                    casual          12.966667 mins
## 2                    member           8.833333 mins
```

```
aggregate(tripdata_v2$ride_length ~ tripdata_v2$member_casual, FUN = max)
```

```
##   tripdata_v2$member_casual tripdata_v2$ride_length
## 1                    casual          34294.067 mins
## 2                    member           1499.933 mins
```

```
aggregate(tripdata_v2$ride_length ~ tripdata_v2$member_casual, FUN = min)
```

```
##   tripdata_v2$member_casual tripdata_v2$ride_length
## 1                    casual                  0 mins
## 2                    member                  0 mins
```

On average, member customers have shorter trips than the casual customers.

The average ride time by each day for members vs casual users.

```
aggregate(tripdata_v2$ride_length ~ tripdata_v2$member_casual + tripdata_v2$weekday, FUN = mean)
```

```
##    tripdata_v2$member_casual tripdata_v2$weekday tripdata_v2$ride_length
## 1                     casual                   1           22.90325 mins
## 2                     member                   1           11.94899 mins
## 3                     casual                   2           19.99747 mins
## 4                     member                   2           11.77548 mins
## 5                     casual                   3           19.21040 mins
## 6                     member                   3           11.88415 mins
## 7                     casual                   4           19.38854 mins
## 8                     member                   4           11.98886 mins
## 9                     casual                   5           20.55960 mins
## 10                    member                   5           12.21362 mins
## 11                    casual                   6           24.08696 mins
## 12                    member                   6           13.74458 mins
## 13                    casual                   7           25.35769 mins
## 14                    member                   7           13.83766 mins
```

Analyze rider data by type and weekday.

```
tripdata_v2 %>%
  mutate(weekday = wday(started_at, label = TRUE)) %>%  #creates weekday field using wday()
  group_by(member_casual, weekday) %>%  #groups by usertype and weekday
  summarise(number_of_rides = n()                           #calculates the number of rides and average duration
  ,average_duration = mean(ride_length_min)) %>%        # calculates the average duration
  arrange(member_casual, weekday)                               # sorts
```

```
## `summarise()` has grouped output by 'member_casual'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 14 × 4
## # Groups:   member_casual [2]
##    member_casual weekday number_of_rides average_duration
##    <chr>         <ord>             <int> <drtn>
##  1 casual        Вс               388011 25.09980 mins
##  2 casual        Пн               277054 22.62878 mins
##  3 casual        Вт               263187 19.63175 mins
##  4 casual        Ср               273823 19.00085 mins
##  5 casual        Чт               308713 19.67468 mins
##  6 casual        Пт               333938 20.53018 mins
##  7 casual        Сб               472082 24.64477 mins
##  8 member        Вс               387117 13.69630 mins
##  9 member        Пн               473249 12.00492 mins
## 10 member        Вт               518507 11.81100 mins
## 11 member        Ср               523770 11.83453 mins
## 12 member        Чт               532154 12.01070 mins
## 13 member        Пт               466985 12.22584 mins
## 14 member        Сб               443169 13.79091 mins
```

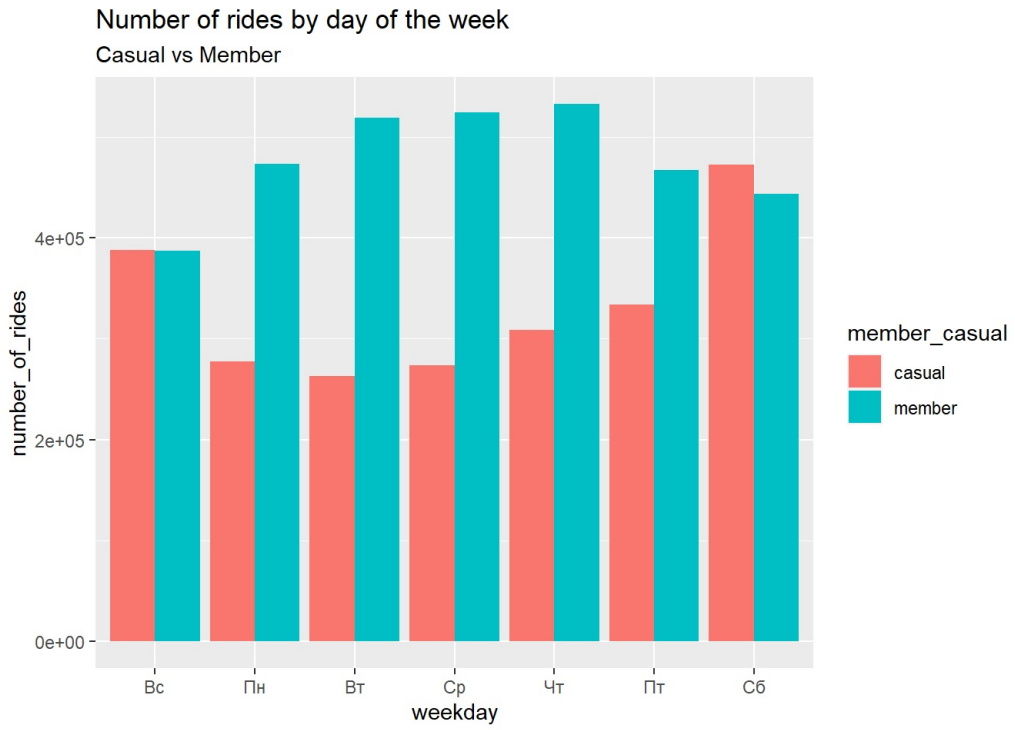
Share

Rides by day of the week

Lets visualize the number of rides by rider type.

```
tripdata_v2 %>%
  mutate(weekday = wday(started_at, label = TRUE)) %>%
  group_by(member_casual, weekday) %>%
  summarise(number_of_rides = n(), average_duration = mean(ride_length_min)) %>%
  arrange(member_casual, weekday)  %>%
  ggplot(aes(x = weekday, y = number_of_rides, fill = member_casual)) +
  geom_col(position = "dodge") +
  labs(title = "Number of rides by day of the week"
       , subtitle = "Casual vs Member")
```

```
## `summarise()` has grouped output by 'member_casual'. You can override using the
## `.groups` argument.
```



Lets create a visualization for average duration.

```
tripdata_v2 %>%
  mutate(weekday = wday(started_at, label = TRUE)) %>%
  group_by(member_casual, weekday) %>%
  summarise(number_of_rides = n(), average_duration = mean(ride_length_min)) %>%
  arrange(member_casual, weekday)  %>%
  ggplot(aes(x = weekday, y = average_duration, fill = member_casual)) +
  geom_col(position = "dodge") +
  labs(title = "Avg duration of rides by day of the week"
       , subtitle = "Casual vs Member")
```

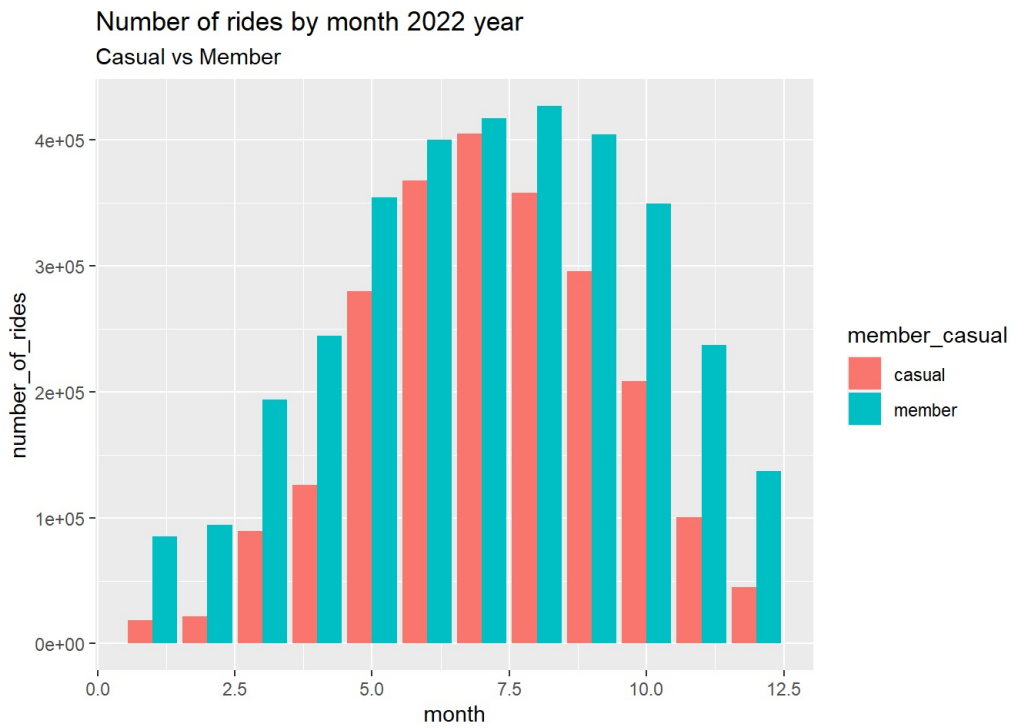
```
## `summarise()` has grouped output by 'member_casual'. You can override using the
## `.groups` argument.
## Don't know how to automatically pick scale for object of type <difftime>.
## Defaulting to continuous.
```

Rides by month

Lets visualize the number of rides by rider type.

```
tripdata_v2 %>%
  group_by(member_casual, month) %>%
  summarise(number_of_rides = n(), average_duration = mean(ride_length_min)) %>%
  arrange(member_casual, month)  %>%
  ggplot(aes(x = month, y = number_of_rides, fill = member_casual)) +
  geom_col(position = "dodge") +
  labs(title = "Number of rides by month 2022 year"
       , subtitle = "Casual vs Member")
```

```
## `summarise()` has grouped output by 'member_casual'. You can override using the
## `.groups` argument.
```



Lets create a visualization for average duration.

```
tripdata_v2 %>%
  group_by(member_casual, month) %>%
  summarise(number_of_rides = n(), average_duration = mean(ride_length_min)) %>%
  arrange(member_casual, month)  %>%
  ggplot(aes(x = month, y = average_duration, fill = member_casual)) +
  geom_col(position = "dodge") +
  labs(title = "Avg duration of rides by month 2022 year"
       , subtitle = "Casual vs Member")
```

```
## `summarise()` has grouped output by 'member_casual'. You can override using the
## `.groups` argument.
## Don't know how to automatically pick scale for object of type <difftime>.
## Defaulting to continuous.
```

Act

- The casual riders take their bikes for longer rides than the members do.
- The casual riders are more likely to use their bikes to ride on weekends.
- The member riders are likely to use their bikes on work days.

For a more in-depth analysis we need more client data such as: client ID in the system, classification of stations depending on the city district (center, cultural area, downtown, countryside), the number and volume of payments, popular destinations for a particular client etc.

Offer clients who get to work an member annual plan, if he or she does not already have one. See who the casual client uses the bike most of the week and let them know about the features and convenience of the annual plan. For more detailed recommendations, I will need a little more information than what was presented.